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Chapter 1 Curves

In this course,

$$\mathbb{R}^n = \{(x_1, x_2, \ldots, x_n) \mid x_i \in \mathbb{R}\}$$

is the Cartesian product of n copies of $\mathbb{R}$.

We shall often denote a point in $\mathbb{R}^2$ as (x, y), and a point in $\mathbb{R}^3$ as (x, y, z).

An interval I is a subset of $\mathbb{R}$ of one of the following forms:

- (a, b), $[a, b)$, $(a, b]$, $[a, b]$;
- $(-\infty, a)$, (a, ∞), $(-\infty, a]$, $[a, \infty)$;
- $(-\infty, \infty)$.

An interval is path-connected and connected.

The Euclidean inner product is the dot product defined as

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

when $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ are vectors in $\mathbb{R}^n$.

The norm of a vector $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ in $\mathbb{R}^n$ is

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{\sum_{i=1}^{n} v_i^2}.$$

The distance between two points $\mathbf{x}_1$ and $\mathbf{x}_2$ in $\mathbb{R}^n$ is

$$d(\mathbf{x}_1, \mathbf{x}_2) = \|\mathbf{x}_1 - \mathbf{x}_2\|.$$

The Cauchy-Schwarz inequality says that

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|.$$

The angle θ between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ is defined as the unique angle between 0° and 180° inclusive such that

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \, \|\mathbf{v}\|}.$$

The two vectors $\mathbf{u}$ and $\mathbf{v}$ are said to be orthogonal or perpendicular if

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0.$$

An open ball of radius $r > 0$ and having center at $\mathbf{x}_0 \in \mathbb{R}^n$ is a subset of $\mathbb{R}^n$ of the form

$$B(\mathbf{x}_0, r) = \{\mathbf{x} \mid \|\mathbf{x} - \mathbf{x}_0\| < r\}.$$

If S is a subset of $\mathbb{R}^n$,

(i) S is open if for every $\mathbf{x}_0$ in S, there exists $r > 0$ such that $B(\mathbf{x}_0, r) \subset S$.

(ii) S is closed if $\mathbb{R}^n \setminus S$ is open.

Every subset S of $\mathbb{R}^n$ divides $\mathbb{R}^n$ into three disjoint regions – the interior of S, denoted by $\operatorname{int} S$, the exterior of S, denoted by $\operatorname{ext} S$, and the boundary of S, denoted by ∂S. If $\mathbf{x}_0$ is a point in $\mathbb{R}^n$,

(i) $\mathbf{x}_0$ is in $\operatorname{int} S$ provided that there exists $r > 0$ such that $B(\mathbf{x}_0, r) \subset S$.

(ii) $\mathbf{x}_0$ is in $\operatorname{ext} S$ provided that there exists $r > 0$ such that $B(\mathbf{x}_0, r) \subset \mathbb{R}^n \setminus S$.

(iii) $\mathbf{x}_0$ is in ∂S provided that for every $r > 0$, $B(\mathbf{x}_0, r)$ intersects S and $\mathbb{R}^n \setminus S$.

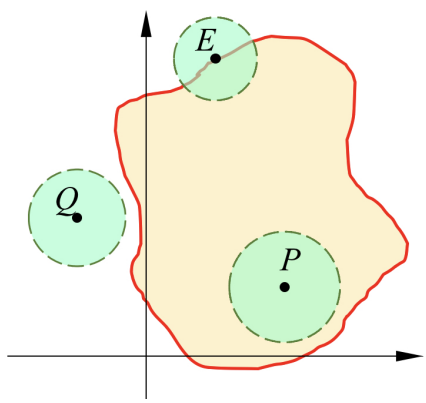


Figure 1.1: Interior, exterior and boundary of a set.

Example 1.1

Find the interior, exterior and boundary of the set

$$S = \left\{(x, y, z) \,|\, x^2 + y^2 + z^2 < 1\right\}.$$

1.1 Curves

Example 1.2 Examples of Curves in $\mathbb{R}^2$

(a) When a and b are constants such that $(a, b) \neq (0, 0)$, $ax + by = c$ is a line in $\mathbb{R}^2$.

(b) When $r > 0$, $x^2 + y^2 = r^2$ is a circle in $\mathbb{R}^2$.

(c) For any real number c, $y = x^2 - c$ is the graph of the function $f(x) = x^2 - c$.

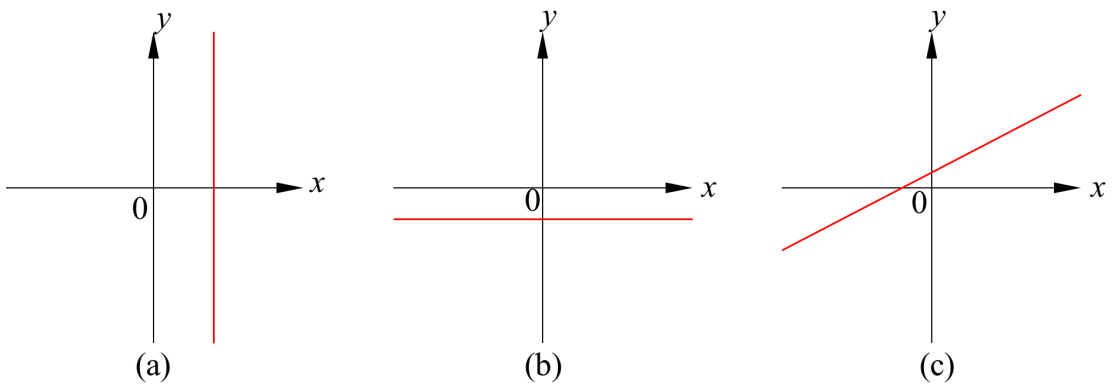


Figure 1.2: (a) A vertical line $x = c$. (b) A horizontal line $y = c$. (c) A general line $ax + by = c$.

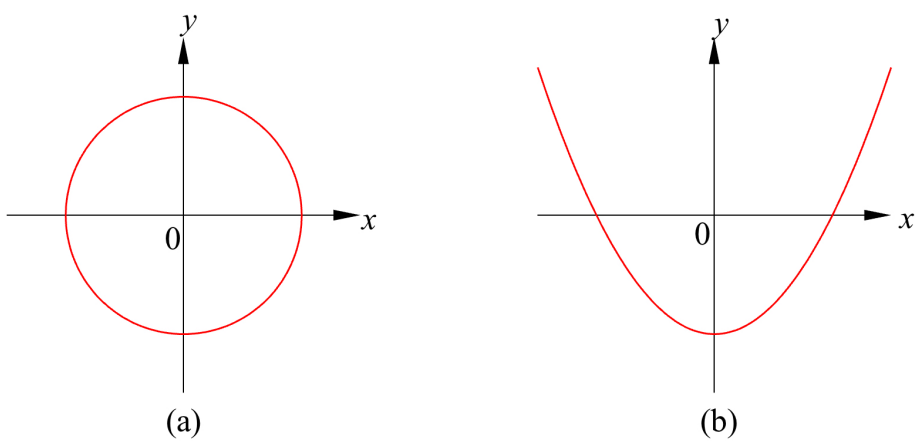


Figure 1.3: (a) A circle $x^2 + y^2 = r^2$. (b) A curve $y = x^2 - c$.

Definition 1.1 Parametrized Curves in $\mathbb{R}^n$

A parametrized curve in $\mathbb{R}^n$ is a continuous function $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ defined on an interval I. Written in components,

$$\boldsymbol{\gamma}(t) = (\gamma_1(t), \gamma_2(t), \ldots, \gamma_n(t)).$$

Definition 1.2 Curves in $\mathbb{R}^n$

The image of a parametrized curve $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ is called a curve in $\mathbb{R}^n$, and $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ is called a parametrization of the curve.

In some textbooks, a parametrized curve is abbreviated as a curve, while the image of a parametrized curve in $\mathbb{R}^n$ is called the trace of the curve.

Example 1.3

When $b \neq 0$, the line $ax+by=c$ is a curve in $\mathbb{R}^2$ with parametrization $\boldsymbol{\gamma}_1 : \mathbb{R} \to \mathbb{R}^2$ given by

$$\boldsymbol{\gamma}_1(t) = \left(t, \frac{c-at}{b}\right).$$

When $a \neq 0$, the line $ax+by=c$ has an alternative parametrization $\boldsymbol{\gamma}_2 : \mathbb{R} \to \mathbb{R}^2$ given by

$$\boldsymbol{\gamma}_2(t) = \left(\frac{c-bt}{a}, t\right).$$

Example 1.4

Let $\boldsymbol{\gamma}_1 : \mathbb{R} \to \mathbb{R}^2$ and $\boldsymbol{\gamma}_2 : \mathbb{R} \to \mathbb{R}^2$ be given respectively by

$$\boldsymbol{\gamma}_1(t) = (t, 1-t), \qquad \boldsymbol{\gamma}_2(t) = (1-t, t).$$

Then they parametrize the same curve.

Example 1.5

A parametrization of the curve $y = x^2$ is given by $\boldsymbol{\gamma}_1 : \mathbb{R} \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}_1(t) = (t, t^2).$$

Alternatively, we can also parametrize the curve by $\boldsymbol{\gamma}_2 : \mathbb{R} \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}_2(t) = (t^2, t^4).$$

Example 1.6

Let r be a positive number. A parametrization of the circle $x^2 + y^2 = r^2$ is given by $\boldsymbol{\gamma}_1 : [0, 2\pi] \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}_1(t) = (r \cos t, r \sin t).$$

The curve parametrized by $\boldsymbol{\gamma}_2 : [-r, r] \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}_2(t) = (t, \sqrt{r^2 - t^2})$$

is not the whole circle $x^2 + y^2 = r^2$. It is only the upper half of the circle. Hence, $\boldsymbol{\gamma}_1 : [0, 2\pi] \to \mathbb{R}^2$ and $\boldsymbol{\gamma}_2 : [-r, r] \to \mathbb{R}^2$ do not parametrize the same curve.

Definition 1.3 Closed Curves

A curve is closed if it has a parametrization $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^n$ with $\boldsymbol{\gamma}(a) = \boldsymbol{\gamma}(b)$.

Example 1.7

The circle $x^2 + y^2 = r^2$ is a closed curve, but the curve $y = x^2$ is not closed.

Definition 1.4 Simple Curves

A curve is simple and not closed if it has a parametrization $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ which is a one-to-one map.

A curve is simple and closed if it has a parametrization $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^n$ which is a one-to-one map on $[a, b)$, and $\boldsymbol{\gamma}(a) = \boldsymbol{\gamma}(b)$.

Example 1.8

The circle $x^2 + y^2 = r^2$ is a simple closed curve. The curve $y = x^2$ is simple but not closed.

Definition 1.5 Smooth Curves

A smooth parametrized curve is a parametrized curve $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ that is C^∞, namely, each components of $\boldsymbol{\gamma}$ is infinitely differentiable.

A curve is smooth if it has a C^∞ parametrization.

It is customary to denote $\dfrac{d\boldsymbol{\gamma}}{dt}$ by $\dot{\boldsymbol{\gamma}}$, $\dfrac{d^2\boldsymbol{\gamma}}{dt^2}$ by $\ddot{\boldsymbol{\gamma}}$, and $\dfrac{d^3\boldsymbol{\gamma}}{dt^3}$ by $\dddot{\boldsymbol{\gamma}}$.

Henceforth, we only consider smooth curves.

Polynomials, rational functions, exponential functions, logarithms, trigonometric functions are smooth on their respective domains.

Composition of smooth functions is smooth.

Definition 1.6 Smooth Closed Curves

A smooth closed curve is a curve that has a parametrization $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^n$ with $\boldsymbol{\gamma}^{(k)}(a) = \boldsymbol{\gamma}^{(k)}(b)$ for all $k \geq 0$.

Example 1.9

Straightlines and circles in $\mathbb{R}^2$ are smooth curves. The curve $y = x^2$ is also smooth.

Example 1.10

Consider the paramtrized curve $\boldsymbol{\gamma} : [0, 2\pi] \to \mathbb{R}^2$ given by

$$\boldsymbol{\gamma}(t) = (\cos^3 t, \sin^3 t).$$

This is a smooth parametrized curve by definition. Eliminating the parameter t, we find that it is defined by the equation

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1.$$

The curve has sharp turns at the four points $(1, 0)$, $(0, 1)$, $(-1, 0)$ and $(0, -1)$. In fact, since

$$\dot{\boldsymbol{\gamma}}(t) = \left(-3\cos^2 t \sin t, 3\sin^2 t \cos t\right),$$

we find that $\dot{\boldsymbol{\gamma}}(t)$ vanishes at these four points.

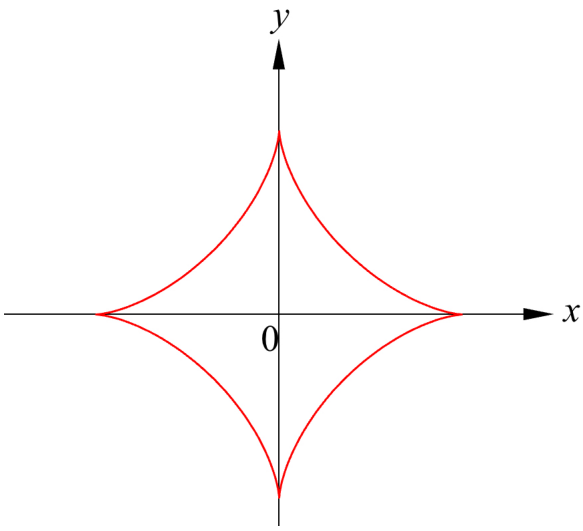


Figure 1.4: The curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1$.

Definition 1.7 Regular Parametrized Curves

A parametrized curve $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ is said to be regular if it is smooth and $\dot{\boldsymbol{\gamma}}(t) \neq \mathbf{0}$ for all $t \in I$.

Example 1.11

The paramtrized curve $\boldsymbol{\gamma} : [0, 2\pi] \to \mathbb{R}^2$ given by

$$\boldsymbol{\gamma}(t) = (\cos^3 t, \sin^3 t)$$

is not regular.

Example 1.12

For the curve $y = x^2$, the parametrization $\boldsymbol{\gamma}_1 : \mathbb{R} \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}_1(t) = (t, t^2)$$

is regular; the parametrization $\boldsymbol{\gamma}_2 : \mathbb{R} \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}_2(t) = (t^2, t^4)$$

is not regular.

From now on, the curves considered in this course are regular smooth curves, unless otherwise specified.

Theorem 1.1 Curves in $\mathbb{R}^2$ that are Defined Implicitly

Let $\mathcal{O}$ be an open subset of $\mathbb{R}^2$, and let $f : \mathcal{O} \to \mathbb{R}$ be a smooth function. Assume that (x_0, y_0) is a point in $\mathcal{O}$ such that

(i) $f(x_0, y_0) = 0$, and

(ii) $\nabla f(x_0, y_0) = (f_x(x_0, y_0), f_y(x_0, y_0)) \neq \mathbf{0}$.

Then there is an open interval I in $\mathbb{R}$, an open subset U of $\mathcal{O}$, and a smooth function $\boldsymbol{\gamma} : I \to \mathbb{R}^2$, $\boldsymbol{\gamma}(t) = (\gamma_1(t), \gamma_2(t))$, such that

(a) $\gamma_1(t) = t$ or $\gamma_2(t) = t$,

(b) for $(x, y) \in U$, $f(x, y) = 0$ if and only if $(x, y) = \boldsymbol{\gamma}(t) = (\gamma_1(t), \gamma_2(t))$.

In other words, if

$$C = \{(x, y) \mid f(x, y) = 0\},$$

then $C \cap U$ is a smooth regular curve parametrized by $\boldsymbol{\gamma} : I \to \mathbb{R}^2$.

Example 1.13

Consider the ellipse $4x^2 + 9y^2 = 36$. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function

$$f(x, y) = 4x^2 + 9y^2 - 36.$$

It is a C^∞ function. If (x_0, y_0) is a point such that $f(x_0, y_0) = 4x_0^2 + 9y_0^2 - 36 = 0$, then $(x_0, y_0) \neq (0, 0)$. It follows that

$$\nabla f(x_0, y_0) = (8x_0, 18y_0) \neq (0, 0).$$

$4x^2 + 9y^2 = 36$ is a smooth curve.

Definition 1.8 Straightlines in $\mathbb{R}^n$

A straightline in $\mathbb{R}^n$ is determined by two points $\mathbf{x}_1$ and $\mathbf{x}_2$ in $\mathbb{R}^n$. It has a regular parametrization given by $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^n$,

$$\boldsymbol{\gamma}(t) = (1-t)\mathbf{x}_1 + t\mathbf{x}_2.$$

A straightline can also be determined by a direction (vector) $\mathbf{v}$ and a point $\mathbf{x}_0$. In this case, it has a regular parametrization given by $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^n$,

$$\boldsymbol{\gamma}(t) = \mathbf{x}_0 + t\mathbf{v}.$$

Definition 1.9 Tangent Vectors and Tangent Lines

Given that $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ is a regular parametrized curve and t_0 is a point in I, the vector $\dot{\boldsymbol{\gamma}}(t_0)$ is the tangent (velocity) vector of the curve at the parameter value t_0, or simply at the point $\mathbf{x}_0 = \boldsymbol{\gamma}(t_0)$ if there is no confusion.

The tangent line to the the curve at the point $\mathbf{x}_0$ is the line with parametrization $\boldsymbol{\ell} : \mathbb{R} \to \mathbb{R}^n$,

$$\boldsymbol{\ell}(t) = \boldsymbol{\gamma}(t_0) + t\dot{\boldsymbol{\gamma}}(t_0).$$

The speed of the curve at the parameter value t_0 is $\|\dot{\boldsymbol{\gamma}}(t_0)\|$.

Example 1.14

At any point on the line

$$\boldsymbol{\gamma}(t) = (1-t)\mathbf{x}_1 + t\mathbf{x}_2,$$

the tangent vector is

$$\dot{\boldsymbol{\gamma}}(t) = \mathbf{x}_2 - \mathbf{x}_1,$$

which is a constant nonzero vector. The tangent line at any point is the line itself.

Example 1.15

If $f : I \to \mathbb{R}$ is a smooth function, its graph is a regular parametrized curve with parametrization $\boldsymbol{\gamma} : I \to \mathbb{R}^2$

$$\boldsymbol{\gamma}(t) = (t, f(t)).$$

At the point $(t, f(t))$, the tangent vector is given by

$$\dot{\boldsymbol{\gamma}}(t) = (1, f'(t)).$$

Example 1.16

In polar coordinates r and θ, a limaçon is a curve of the form $r = b + a\cos\theta$. It is a closed curve that can be parametrized by $\boldsymbol{\gamma} : [0, 2\pi] \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}(t) = ((b + a\cos t)\cos t, (b + a\cos t)\sin t)\,.$$

When $|a| > |b|$, it has self-intersection at the point $(0, 0)$. For example, when $b = 1$ and $a = 2$, $\boldsymbol{\gamma}(t)$ is the point $(0, 0)$ when $t = \pm 2\pi/3$. However, the tangent vectors are different.

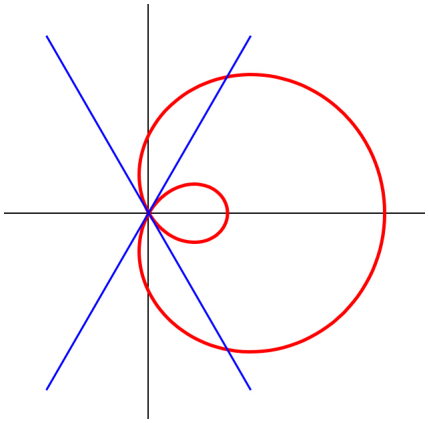


Figure 1.5: The limaçon $r = 1 + 2\cos\theta$.

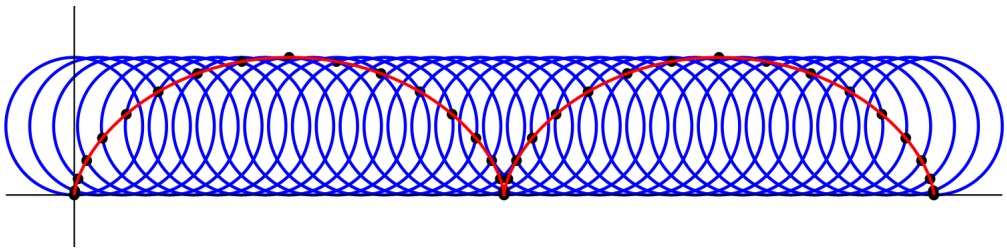
Example 1.17

A cycloid is the plane curve traced out by a point on the circumference of a circle as it rolls without slipping along a straightline.

(a) Show that, if the straightline is the x-axis and the circle has radius $a > 0$, the cycloid can be parametrized as $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}(t) = (a(t - \sin t), a(1 - \cos t)) .$$

(b) Find the tangent vector at each point of the cycloid.



Definition 1.10 Normal Lines

The normal line to a curve in $\mathbb{R}^2$ at a point $\mathbf{x}_0$ is the straightline passing through $\mathbf{x}_0$ and perpendicular to the tangent line at $\mathbf{x}_0$.

Example 1.18

Find the tangent and normal lines to the curve

$$\boldsymbol{\gamma}(t) = (2\cos t - \cos 2t, 2\sin t - \sin 2t)$$

at the point corresponding to $t = \dfrac{\pi}{4}$.

1.2 Arc-Length

Definition 1.11 Arc-Length

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ be a smooth curve and let $\mathbf{u} = \boldsymbol{\gamma}(a)$ and $\mathbf{v} = \boldsymbol{\gamma}(b)$ be two points on the curve with $a < b$. The arc-length of the curve from the point $\mathbf{u}$ to $\mathbf{v}$ is given by

$$\int_a^b \|\dot{\boldsymbol{\gamma}}(t)\| dt.$$

The arc-length function of the curve is the function $s : I \to \mathbb{R}$,

$$s(t) = \int_{t_0}^t \|\dot{\boldsymbol{\gamma}}(\tau)\| d\tau,$$

where t_0 is a point in I. If a different t_0 is chosen, the resulting arc-length function differs by a constant.

Example 1.19

Find the arc-length function of the helix whose parametrization is given by $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^3$,

$$\boldsymbol{\gamma}(t) = (a\cos t, a\sin t, bt),$$

where a and b are nonzero constants.

Proposition 1.2

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ be a smooth regular curve and let $s : I \to \mathbb{R}$ be its arc-length function. Then

(i) $s : I \to \mathbb{R}$ is a smooth function and

$$\frac{ds}{dt} = \|\dot{\boldsymbol{\gamma}}(t)\|$$

is the speed function.

(ii) $s : I \to \mathbb{R}$ is a strictly increasing.

Proposition 1.3

If $\mathbf{u} : I \to \mathbb{R}^n$ is a smooth function, then

$$\left\langle \frac{d\mathbf{u}}{dt}(t), \mathbf{u}(t) \right\rangle = 0 \qquad \text{for all } t \in I,$$

if and only if the vector $\mathbf{u}(t)$ has constant norm.

Example 1.20

If $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ is a parametrized curve with constant speed, then $\ddot{\boldsymbol{\gamma}}(t)$ is perpendicular to $\dot{\boldsymbol{\gamma}}(t)$ for all $t \in I$.

Theorem 1.4

Given any two distinct points $\mathbf{p}$ and $\mathbf{q}$ in $\mathbb{R}^n$, if $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^n$ is any regular parametrized curve with $\boldsymbol{\gamma}(a) = \mathbf{p}$ and $\boldsymbol{\gamma}(b) = \mathbf{q}$, then

$$\int_a^b \|\dot{\boldsymbol{\gamma}}(t)\| dt \geq \|\mathbf{q} - \mathbf{p}\|,$$

and the equality holds if and only if the image of $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^n$ is the straightline segment between $\mathbf{p}$ and $\mathbf{q}$.

This theorem says that the straightline is the shortest curve that joins $\mathbf{p}$ to $\mathbf{q}$.

1.3 Reparametrization

The straightline segment between two points $\mathbf{p}$ and $\mathbf{q}$ in $\mathbb{R}^n$ has a regular parametrization given by $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^n$,

$$\boldsymbol{\gamma}(t) = \mathbf{p} + \phi(t)(\mathbf{q} - \mathbf{p}),$$

where $\phi : [a, b] \to \mathbb{R}$ is any strictly increasing smooth function with

$$\phi(a) = 0 \quad \text{and} \quad \phi(b) = 1.$$

The arc-length of this line segment is $\|\mathbf{q} - \mathbf{p}\|$, independent of the parametrization.

Definition 1.12 Reparametrization

Given the parametrized smooth curves $\boldsymbol{\gamma}_1 : I_1 \to \mathbb{R}^n$ and $\boldsymbol{\gamma}_2 : I_2 \to \mathbb{R}^n$, we say that $\boldsymbol{\gamma}_2 : I_2 \to \mathbb{R}^n$ is a reparametrization of $\boldsymbol{\gamma}_1 : I_1 \to \mathbb{R}^n$ if there exists an increasing smooth bijection $\phi : I_1 \to I_2$ such that $\phi^{-1} : I_2 \to I_1$ is also smooth, and

$$\boldsymbol{\gamma}_1(t) = \boldsymbol{\gamma}_2(\phi(t)).$$

An increasing bijection $\phi : I_1 \to I_2$ is necessary strictly increasing. The inverse $\phi^{-1} : I_2 \to I_1$ is also an increasing bijection.

Therefore, if $\boldsymbol{\gamma}_2 : I_2 \to \mathbb{R}^n$ is a reparametrization of $\boldsymbol{\gamma}_1 : I \to \mathbb{R}^n$, then $\boldsymbol{\gamma}_1 : I_1 \to \mathbb{R}^n$ is a reparametrization of $\boldsymbol{\gamma}_2 : I_2 \to \mathbb{R}^n$. Hence, in this case, we say that $\boldsymbol{\gamma}_1 : I_1 \to \mathbb{R}^n$ and $\boldsymbol{\gamma}_2 : I_2 \to \mathbb{R}^n$ are equivalent parametrizations of the same curve.

By chain rule,

$$(\phi^{-1})'(\phi(t))\phi'(t) = 1 \qquad \text{for all } t \in I_1.$$

Hence, $\phi'(t) \neq 0$ for all $t \in I_1$. Since

$$\boldsymbol{\gamma}_1'(t) = \boldsymbol{\gamma}_2'(\phi(t))\phi'(t) \qquad \text{for all } t \in I_1,$$

if $\boldsymbol{\gamma}_1 : I_1 \to \mathbb{R}^n$ is a regular smooth curve, then $\boldsymbol{\gamma}_2 : I_2 \to \mathbb{R}^n$ is also a regular smooth curve.

Example 1.21

Let $\boldsymbol{\gamma}_1 : [0, 2\pi] \to \mathbb{R}^2$ and $\boldsymbol{\gamma}_2 : [-\frac{\pi}{2}, \frac{3\pi}{2}] \to \mathbb{R}^2$ be parametrized curves given respectively by

$$\boldsymbol{\gamma}_1(t) = (\cos t, \sin t), \qquad \boldsymbol{\gamma}_2(t) = (-\sin t, \cos t).$$

Show that they are equivalent parametrization of the same smooth curve.

Definition 1.13 Parametrization by Arc-Length

Let I be an interval and let $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ be a smooth parametrized curve. We say that $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ is a parametrization by arc-length if

$$\|\dot{\boldsymbol{\gamma}}(t)\| = 1 \qquad \text{for all } t \in I.$$

Equivalently, we say that $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ is a unit speed parametrization.

A parametrization by arc-length is necessarily regular. If $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ is a parametrization of a curve C by arc-length, it is customary to denote the parameter by s. For any s_1 and s_2 in I, the arc-length of the curve from $\boldsymbol{\gamma}(s_1)$ to $\boldsymbol{\gamma}(s_2)$ is $s_2 - s_1$.

If $\boldsymbol{\gamma}_1 : I_1 \to \mathbb{R}^n$ is a reparametrization of $\boldsymbol{\gamma} : I \to \mathbb{R}^n$, and both $\boldsymbol{\gamma}_1 : I_1 \to \mathbb{R}^n$ and $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ are parametrizations by arc-length, then there is a constant c such that

$$\boldsymbol{\gamma}_1(s) = \boldsymbol{\gamma}(s + c) \qquad \text{for all } s \in I_1.$$

Example 1.22

The parametrization $\boldsymbol{\gamma} : [0, 2\pi] \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}(s) = (\cos s, \sin s)$$

parametrizes the unit circle $x^2 + y^2 = 1$ by arc-length.

Example 1.23

Let $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^3$ be the parametrized curve given by

$$\boldsymbol{\gamma}(t) = (3\cos t, 3\sin t, 4t)\,.$$

Find a reparametrization of the curve by arc-length.

Example 1.24

Let $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^3$ be the parametrized curve given by

$$\boldsymbol{\gamma}(t) = \left(t, t^2, t^3\right).$$

Then

$$\dot{\boldsymbol{\gamma}}(t) = \left(1, 2t, 3t^2\right).$$

Hence, the arc-length function is

$$s(t) = \int_0^t \sqrt{1 + 4\tau^2 + 9\tau^4}\, d\tau.$$

In principle, the curve has a reparametrization by arc-length, but we are unable to write it out explicitly.

Theorem 1.5

If $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ is a regular curve, then it has a reparametrization by arc-length.

1.4 Cross Product, Orthogonal Matrices and Isometries of $\mathbb{R}^n$

Recall that if $\mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{w} = (w_1, w_2, w_3)$ are two vectors in $\mathbb{R}^3$, their cross product is defined as

$$\mathbf{v} \times \mathbf{w} = \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix},$$

where $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$, $\mathbf{k} = (0, 0, 1)$. The vector $\mathbf{v} \times \mathbf{w}$ is perpendicular to both $\mathbf{v}$ and $\mathbf{w}$, and has magnitude

$$\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\| \, \|\mathbf{w}\| \, \sin\theta,$$

where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$.

If $\mathbf{v}$ and $\mathbf{w}$ are parallel, $\mathbf{v} \times \mathbf{w} = \mathbf{0}$.

If $\mathbf{v}$ and $\mathbf{w}$ are not parallel, the vectors $\mathbf{v}, \mathbf{w}$ and $\mathbf{v} \times \mathbf{w}$ form a right-handed system.

If $\mathbf{v}$ and $\mathbf{w}$ are perpendicular,

$$\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\| \, \|\mathbf{w}\|.$$

If $\mathbf{u} = (u_1, u_2, u_3)$, $\mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{w} = (w_1, w_2, w_3)$ are vectors in $\mathbb{R}^3$, then

$$\langle \mathbf{u}, \mathbf{v} \times \mathbf{w} \rangle = \det \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}.$$

Definition 1.14 Orthogonal Matrices

An $n \times n$ matrix $\mathbf{P}$ is orthogonal if

$$\mathbf{P}\mathbf{P}^T = \mathbf{I}_n.$$

An $n \times n$ matrix $\mathbf{P}$ is orthogonal if and only if the column vectors of $\mathbf{P}$ form an orthonormal basis of $\mathbb{R}^n$. Namely, if

$$\mathbf{P} = \left[\mathbf{v}_1 \,\middle|\, \mathbf{v}_2 \,\middle|\, \cdots \,\middle|\, \mathbf{v}_n\right],$$

then $\mathbf{P}$ is orthogonal if and only if $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an orthonormal basis of $\mathbb{R}^n$.

The determinant of an orthogonal matrix $\mathbf{P}$ can only be 1 or -1.

If

$$\mathbf{P} = \left[\mathbf{v}_1 \,\middle|\, \mathbf{v}_2 \,\middle|\, \mathbf{v}_3\right]$$

is an 3×3 orthogonal matrix, $\det \mathbf{P} = 1$ if and only if $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a right-handed triplet.

Definition 1.15 Orthogonal Transformations

An orthogonal transformation of $\mathbb{R}^n$ is a linear transformation $\mathbf{T} : \mathbb{R}^n \to \mathbb{R}^n$ defined by a $n \times n$ orthogonal matrix $\mathbf{P}$. Namely,

$$\mathbf{T}(\mathbf{x}) = \mathbf{P}\mathbf{x}.$$

Proposition 1.6

Let $\mathbf{T} : \mathbb{R}^n \to \mathbb{R}^n$, $\mathbf{T}(\mathbf{x}) = \mathbf{P}\mathbf{x}$ be an orthogonal transformation defined by the orthogonal matrix $\mathbf{P}$. Then for any $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$,

$$\langle \mathbf{T}(\mathbf{x}), \mathbf{T}(\mathbf{y}) \rangle = \langle \mathbf{x}, \mathbf{y} \rangle.$$

This implies that an orthogonal transformation preserves lengths and angles.

Definition 1.16 Isometry of $\mathbb{R}^n$

A map $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ is an isometry if for any two points $\mathbf{u}$ and $\mathbf{w}$ in $\mathbb{R}^n$,

$$d(\mathbf{F}(\mathbf{u}), \mathbf{F}(\mathbf{w})) = d(\mathbf{u}, \mathbf{w}).$$

Example 1.25

Let $\mathbf{a}$ be a point in $\mathbb{R}^n$. The translation $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ defined by

$$\mathbf{F}(\mathbf{x}) = \mathbf{x} + \mathbf{a}$$

is an isometry.

Example 1.26

An orthogonal transformation is an isometry.

Proposition 1.7

If $\mathbf{F}_1 : \mathbb{R}^n \to \mathbb{R}^n$ and $\mathbf{F}_2 : \mathbb{R}^n \to \mathbb{R}^n$ are isometries, then $(\mathbf{F}_2 \circ \mathbf{F}_1) : \mathbb{R}^n \to \mathbb{R}^n$ is also an isometry.

Corollary 1.8

Let $\mathbf{a}$ be a point in $\mathbb{R}^n$, and let $\mathbf{P}$ be an $n \times n$ orthogonal matrix. The map $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ defined by

$$\mathbf{F}(\mathbf{x}) = \mathbf{P}\mathbf{x} + \mathbf{a}$$

is an isometry.

Theorem 1.9

If $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ is an isometry and $\mathbf{F}(\mathbf{0}) = \mathbf{0}$, $\mathbf{F}(\mathbf{e}_i) = \mathbf{e}_i$ for $1 \leq i \leq n$, then $\mathbf{F}(\mathbf{x}) = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$.

Theorem 1.10

If $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ is an isometry, there is a point $\mathbf{a}$ in $\mathbb{R}^n$, and an $n \times n$ orthogonal matrix $\mathbf{P}$ such that

$$\mathbf{F}(\mathbf{x}) = \mathbf{P}\mathbf{x} + \mathbf{a}.$$

Corollary 1.11

A map $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ is an isometry if and only if there is a point $\mathbf{a}$ in $\mathbb{R}^n$, and an $n \times n$ orthogonal matrix $\mathbf{P}$ such that

$$\mathbf{F}(\mathbf{x}) = \mathbf{P}\mathbf{x} + \mathbf{a}.$$

Remark 1.1 Direct Isometry

An isometry $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$,

$$\mathbf{F}(\mathbf{x}) = \mathbf{P}\mathbf{x} + \mathbf{a}$$

defined by the orthogonal matrix $\mathbf{P}$ is called a direct isometry if $\det \mathbf{P} = 1$.

Proposition 1.12

Let $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ be vectors in $\mathbb{R}^3$. Then

$$\langle \mathbf{a} \times \mathbf{b}, \mathbf{c} \times \mathbf{d} \rangle = \det \begin{bmatrix} \langle \mathbf{a}, \mathbf{c} \rangle & \langle \mathbf{a}, \mathbf{d} \rangle \\ \langle \mathbf{b}, \mathbf{c} \rangle & \langle \mathbf{b}, \mathbf{d} \rangle \end{bmatrix}.$$

Proposition 1.13

If $\mathbf{v}$ and $\mathbf{w}$ are vectors in $\mathbb{R}^3$, $\mathbf{P}$ is a 3×3 orthogonal matrix with $\det \mathbf{P} = 1$, then

$$(\mathbf{P}\mathbf{v}) \times (\mathbf{P}\mathbf{w}) = \mathbf{P}(\mathbf{v} \times \mathbf{w}).$$